

EDUCATION

- **University of St. Andrews** United Kingdom
MSc Statistics *Distinction, Deans' List, grade 17.9/20, September 2022*
 - **Modules:** Markov Chains, Bayesian Statistics, Multivariate Analysis, Statistical Modelling using GLM/GAMs, Quantitative risk analysis, Machine Learning
 - **Dissertation:** An Ensemble of HMM and KNN for signal generation in horse betting markets. Grade 19.3/20 - Highest grade in my cohort
- **University of Sheffield** Remote, United Kingdom
Graduate Certificate in Statistics *Distinction, grade 90/100, July 2021*
 - **Modules:** Mathematical Methods for Statistics, Probability and Probability Distributions, Basic Statistics
- **Politechnica University of Timisoara** Timisoara, Romania
B.S.E in Computer Science *July, 2014*
 - **Modules:** Software Engineering, Computer Architecture, Algorithms, Software Development Fundamentals, Design Patterns, Programming Languages, Databases

EXPERIENCE

- **Citadel** London, United Kingdom
Senior Software Engineer *April 2023 - Present*
 - **Global Fixed Income - Data:** Infrastructure team that is in charge of both real-time and reference data for the entire department. Kafka, Airflow, Python, C++, gRPC, Data Pipeline, Redis
 - * **Real-time tick pipeline** that handles 13 billion ticks a day
 - * Decreased in process performance time by 80% using perf flamegraph.
 - * Decreased memory footprint of each worker in the cluster from 15GB RAM to 1 GB by optimizing based on sharded scope.
 - * **Reference data** scraping, parsing and management
 - * Increased read APIs fault tolerance by implementing retry mechanisms around database connectivity which reduced to 0 support tickets.
- **Bloomberg LP** London, United Kingdom
Senior Software Engineer *2019 - September 2021*
 - **Taxonomy and Metadata System:** Infrastructure team developing a taxonomy and metadata system to support various asset classes at Bloomberg. Python, GraphQL, Redis, RabbitMQ, Solr, RDF, PostgreSQL, Data Pipeline
 - * **Architected ingestion pipelines** that handle ingestion of metadata for datasets such as commodity index tickers.
 - * Added **support for sharding** to increase the speed of the ingestion pipeline. Split up the initial ingestion into 8 shards that are running distributed across an entire cluster.
 - * Implemented node overlay and concatenation executed on RDF graphs.
 - * Worked on **rearchitecting the ingestion pipeline** from a synchronous pipeline to a file based ingestion.
 - * Migrated the codebase from Python 2 to Python 3 **decreasing the overall response time of services by 15%**
 - * Implemented services in GraphQL to avoid overfetching and underfetching on the existing Bloomberg specific microservice framework.
 - * Led **integration of the metadata system with timeseries datasets**, a company wide effort to support non-tickerized data.
 - * Worked on a side project, debugging tool, to help find metadata for given entities regardless of the source of provenience, using a **topological sort algorithm on service APIs and RDF Graphs**.
 - * Led the **research into adoption of triplestores** at Bloomberg, an effort across 6 departments towards choosing the right solution to cover the usecases presented at Bloomberg. Researched on solutions such as Virtuoso or TerminusDB.

- * Organized weekly department-wide design discussions, to share knowledge regarding ongoing work.
- * Team lead backup for a team of 7 engineers.

• Bloomberg LP

London, United Kingdom

Software Engineer

2016 - 2019

- **Real Time Analytics:** Application team, maintaining a large variety of analytics, integrated in different products across Bloomberg. Processing several billion trades per day. C++, Distributed, Realtime
 - * Implemented Average Volume at Time for non equities and Top Exchanges in the past X minutes.
 - * Implemented a **load testing facility** that is used to simulate the impact of market open on the system to assess performance and capacity. I identified a bottleneck in the system and by fixing it, subscriptions are being processed 3 times faster.
 - * **Increased availability** of an analytic by implementing fallback to disk in case the database connection cannot be established. This prevents the 1 minute downtime per week when the database cluster is being rebooted.
- **Recommendation Disclosures and Trade Ideas:** Python, Typescript, Microservices, Event Sourcing
 - * Implemented asynchronous messaging using RabbitMQ to allow distribution of recommendations. Added support to publish messages, implemented consumers and set up the middleware.
 - * Reduced the codebase by 8% by removing around 100.000 lines of dead or unused code.

• Intelligent Software Systems

Timisoara, Romania

Software Engineer

2013 - 2016

- Implemented a synchronization tool between a desktop application and an external CRM system (workbooks.com)
- Improved the FAFSA (Federal Student Aid) questionnaire by fixing bugs and adding extra features.

SKILLS

• Coding

- **attrs-strict** <https://github.com/bloomberg/attrs-strict>: Created and open-sourced a library that allows runtime validation for attrs dataclasses in Python.
- **bloomberg-bas-middleware**: Maintainer and creator of an inner source library used for Python services at Bloomberg. Created Flask-like middlewares that are used in over 10.000 services running in Python 3 at Bloomberg. Any new service has by default the functionality opted-in.
- **Python Guild**: Part of a group of 18 engineers leading the direction and adoption of Python at Bloomberg, led the Design Reviews Working group, helping teams to bring projects to a broader audience.
- **ML 101**: Participated in a 6 month hands on Machine Learning training. The timeseries clustering project developed in a team of 3 was one of the 2 projects out of 15 to be showcased in front of senior management at Bloomberg.

• Public speaking for large audiences

- **"How to migrate services to Python 3"** - Attendance of over 200 engineers.
- **"Modeling the commodity world"** - Presentation on the architecture of the system I worked on and the different architectural decisions taken in the past years. Attendance of over 150 engineers.

• Mentorship

- Mentored in workshops to help engineers migrate their services to Python 3.
- Mentored an intern during the summer internship program.
- Mentored new joiners in the team and helped people contribute to inner-source projects at Bloomberg

• Architecture and technology: Large scale distributed systems, Data pipelines, Microservices, GraphQL, PostgreSQL, Redis, Solr, RabbitMQ, Event Sourcing, Git, RDF

• Languages: Python, Javascript, Typescript, SQL, C#

• Mathematics: Probability theory, Markov chains, Bayesian models, GLMs, statistical inference

• Machine Learning: Regression, Trees, Ensemble Methods, Boosting, Neural networks, unsupervised learning

• Languages spoken: I speak English(IELTS Grade 8.0) and Romanian fluently. Currently living in London.

• Citizenships: British, Romanian, Hungarian

PERSONAL PROJECTS

- **G-Research speed coding challenge:** (October 2021) Second place in Scotland at the university coding challenge organised by G-Research. Competed against teams of 4.
- **Cryptocurrency analysis** <https://github.com/erikseulean/cryptoscraping>: (January 2018) Scraped cryptocurrency data from different sources and created a library that allows to easily analyse prices, market cap and volume. I implemented and plotted the Ichimoku indicator and created correlation plots between google trends and different cryptocurrencies. Python: Numpy, Pandas, Matplotlib, Multiprocessing, Jupyter
- **ColorBlocks:** (November 2015) Awarded 2nd prize competing against 12 teams at HackTM, the largest Hackathon in Eastern Europe lasting 50 hours straight. I was part of a team of three and we developed a cross-platform puzzle game for mobile and tablet devices. Available on Google Play as ColorBlocks. LUA
- **Cycling Community Platform:** (October 2012) Awarded 2nd prize competing against 20 teams in Microsoft Excite, three week competition. Developed a mobile and web based platform for the local cycling community. I was part of a team of three students. C#.NET, Javascript, SQL